MILITARY SPECIFICATION

CABLE, SPECIAL PURPOSE, RADIO FREQUENCY, BUOYANT, COAXIAL

TYPE RG-384/U

1. SCOPE

1.1. This specification covers the design and construction of a special purpose radio frequency buoyant coaxial cable.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-17	-	Cables, Radio Frequency: Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead
MIL-C-12000	-	Cable, Cord and Wire, Electric; Packaging of
MIL-I-45208	-	Inspection System Requirements

FEDERAL

L-P-390	-	Plastic Molding Material, Polyethylene, Low and Medium Density
QQ-W-343	-	Wire, Electrical (Uninsulated)

STANDARDS

MILITARY

AN 6227	-	Packing "O" Ring Hydraulic

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York, New York 10016.)

USN UNDERWATER SOUND LABORATORY (USNUSL)

USNUSL Test Gland Assembly Unit Number 44500B
USNUSL Technical Memorandum Number 220-74-62

(Application for copies should be addressed to the U. S. Navy Underwater Sound Laboratory, Fort Trumbull, New London, Connecticut 06320.)

3. REQUIREMENTS

3.1 First article sample. - Prior to beginning production, a sample shall be tested as specified in 4.4 (see 6.2).

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3.2 Mechanical characteristics. -

3.2.1 Specific gravity. - The cable shall have the lowest possible overall specific gravity consistent with the requirement specified herein. The specific gravity shall not exceed .92 and specific gravity at atmospheric pressure in fresh water shall be not less than .75 and not greater than .84 when tested as specified in 4.6.4.

3.2.2 Cold bend. - The cable shall pass the cold bend requirements of 4.6.5.

3.2.3 Crack resistance. - The cable shall be resistant to stress cracking when tested as specified in 4.6.6.

3.2.4 Watertightness. - The cable core, up to and including the sheath of 3.3.5, shall be capable of withstanding a hydrostatic pressure of 600 pounds per square inch gage (psig) applied to the open end of a 5-foot specimen employing dash no. 19 of USNUSL Test Gland Assembly Unit Number 44500B for a period of 2 hours with no leakage (see 3.3.4).

3.2.5 Concentricity. - The coaxial center core of the cable shall be concentric with the cable jacket with a maximum allowable eccentricity of 1/32 inch between the coaxial core and the jacket.

3.3 Construction. -

3.3.1 Center conductor. - The center conductor of the cable shall be No. 16 AWG, Type S drawn and fully annealed, bare copper wire, of continuous length with no joint or splice in accordance with QQ-W-343.

3.3.2 Solid core dielectric. - The solid core dielectric shall consist of Type II, Class L, Grade 7A polyethylene as specified in L-P-390 having an outside diameter of 0.180 $\pm$ .004 inch. The surface of the dielectric shall be smooth so that an O-ring of .070 $\pm$ .003 inch cross section and .145 $\pm$ .005 inch I.D. in accordance with dash no. 2 of AN6227 will seal on the dielectric without any leakage at any pressure from 0 to 600 psig.

3.3.3 Outer conductor. - The outer conductor braid shall conform to the requirements of MIL-C-17 except that it shall consist of braided flat copper ribbon .050 $\pm$ .005 inch wide and .002 $\pm$ .0003 inch thick, employing 16 carriers and 6 picks per inch.

3.3.4 Waterproofing. - A suitable water blocking compound, Vistanex LM-MS or equal, shall be used to seal the braid without causing deterioration of electrical characteristics. Any waterproofing materials shall be easily removed without special procedures so that connectors can be easily assembled onto the cable and an O-ring seal made on the dielectric.

3.3.5 Sheath. - The outer conductor and waterblocking compound shall be covered with a thin continuous polyethylene sheath. The O.D. of this sheath shall be .220 $\pm$ .005. The surface of the sheath shall be smooth so that an O-ring of .070 $\pm$ .003 cross section and .208 $\pm$ .005 I.D. in accordance with dash no. 4 of AN6227 will seal on the sheath without any leakage at any pressure from 0 to 600 psig.

3.3.6 Strength members. - The strength members shall consist of at least twenty strands of .038 inch diameter fiberglass, ECC 75-5/3 Latex 2.0 "S" coated, laid on the sheath with a minimum left hand lay of 20 inches.

3.3.7 Jacket. - The jacket shall consist of two layers of black foamed polyethylene. The extreme diameter of the cable shall be 0.650 $\pm$ .025 inch. The surface of the jacket shall be smooth so that an O-ring of 0.360 inch cross section and 0.620 I.D. (Federal stock #H-5330-064-6585) will seal anywhere along the entire length of the cable with zero leakage at any pressure from 0 to 600 psig when applied for 2 hours. The surface of the jacket shall be of uniform hardness and free of major imperfections such as blow holes, cuts, valleys, bruises, and abrupt changes of diameter within the tolerance range. Abrupt changes are defined as variations in diameter greater than .005 inch per inch of length.

3.4 Strength. -

3.4.1 Cable strength. - Minimum strength of the cable shall be 2000 pounds. The cable strength shall be determined by means of a power driven tensile machine. The rate of travel of the power actuated grip

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shall be adjusted to move at a rate of 12 + 2 feet per minute. Distance between grips shall be 5 feet prior to the application of the load. A reduction of the jacket diameter to .580 inch shall be considered a failure.

3.4.2 Shear strength. - The shear strength between strength members and jacket shall be a minimum of 100 pounds per linear foot of cable. A suitable device such as a Kellems grip shall be used to facilitate the measurement of the shear strength. At least two 1-foot specimens shall be tested. All specimens tested shall pass this test.

3.5 Length. - The cable shall be provided in not less than 1700 feet and not more than 1800 feet continuous lengths with no joint or splice.

3.6 Identification. - Manufacturers identification shall consist of coloring the solid dielectric core in accordance with a manufacturer's identification color to be assigned by the command or agency concerned. Coloring of the core shall not affect the characteristics of the cable. There shall be no markings on the outside surface of the cable.

3.7 Resistance. -

3.7.1 DC resistance. - When tested in accordance with 4.6.7, the DC resistance of the outer conductor shall not be greater than 7.0 ohms per 1000 feet.

3.7.2 Resistance at 20 kHz. - When tested in accordance with 4.6.8, the resistance at 20 kHz shall not be greater than 15 ohms per 1000 feet.

3.8 Attenuation. - When tested in accordance with 4.6.9, the attenuation at 2, 10, and 30 MHz shall not be greater than .45, .90, and 1.6 dB per 100 feet, respectively.

3.9 Diameter recordings. - Diameter recordings of each reel of cable shall be supplied as specified in 4.6.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality control system. - The supplier shall provide and maintain a quality control system in accordance with MIL-I-45208.

4.2 Inspection conditions. - Unless otherwise specified herein, all inspections shall be performed at room ambient temperature, pressure and humidity.

4.3 Classification of inspection. - The method of examination and testing the equipment fall within the following classifications:

(a) First article inspection (see 4.4)
(b) Quality conformance inspection (see 4.5)

4.4 First article inspection. - Unless otherwise specified (see 6.2), a 300-foot sample of cable shall be required for first article inspection. First article inspection shall consist of all examinations and testing necessary to determine compliance with the requirements of this specification (see 4.5.2 and 4.6).

4.4.1 First article sample inspection. - A report of the tests conducted and a 100-foot length taken from the 300-foot cable sample shall be forwarded to the Naval Ship Engineering Center, Philadelphia Division, Submarine Antenna Quality Assurance Department (SAQAD), Code 6700D. Unless otherwise instructed, the first article inspection shall be conducted at SAQAD and the test report shall be forwarded to the Naval Electronic Systems Command, Code 05143 (information copy), the contracting officer and the

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contractor. The manufacturer shall not initiate production prior to receipt of approval by the contracting officer except at his own risk.

4.5 Quality conformance inspection. -

4.5.1 Sampling. - A sample of cable shall be selected from each reel for the examination and tests of 4.5.2 and 4.6.4. Samples of sufficient length shall be selected from every 40,000 feet of cable for the tests of 4.6.2, 4.6.3, 4.6.5, 4.6.6, 4.6.7, 4.6.8, 4.6.9 and , 4.6.10.

4.5.2 Visual and dimensional examination. - The samples selected in accordance with 4.5.1 shall be examined to verify that the design, construction, physical dimensions, and workmanship are in accordance with this specification. The sample shall also be examined to verify that it meets the concentricity requirements of 3.2.5.

4.6 Test procedures. -

4.6.1 Dimensions and out-of-round. - The overall diameter of the outer jacket shall be measured. The cable making equipment shall be fitted with electronic or mechanical measuring device(s) to produce graphic records of the cable produced. Two recordings shall be made as nearly simultaneously as possible of each diameter, 90 degrees apart, and at a point in manufacturing where further dimensional changes such as cooling, tension, and so forth will not occur. The recordings shall be permanent and reproducible by a common commercial process. The recording accuracy shall be no less than +5 percent of the tolerance being measured, and any portion of the recording of the O. D. shall be identifiable to within ±2 feet of the cable being measured and not less than 1/20 of an inch of paper per each 1 foot of cable being monitored. Each recording shall be identified with the reel of cable being monitored. The beginning of the cable length and end of the cable run shall be indicated on the recordings.

4.6.1.1 Calibration. - The zero position on the graph and an indication of which side of zero is over and which side is under the nominal cable diameters shall be required, and an indication of specific values in thousandths of an inch shall be made on the recording with regard to the dielectric and outer jacket outside diameters.

4.6.2 Breaking strength. - The breaking strength of the cable shall be determined by means of a power-driven tensile machine as specified in 3.4.1.

4.6.3 Shear strength. - A suitable device shall be constructed to measure the shear strength in accordance with 3.4.2.

4.6.4 Specific gravity. - The specific gravity shall not exceed .92 when measured under a hydrostatic pressure of 600 psig and at room temperature in fresh water. The measurement shall be made after a continuous immersion time of two hours at 600 psig. The cable sample shall not be removed from the pressure tank nor the pressure reduced until after the measurement is completed. An approved test method is contained in USNUSL Technical Memorandum No. 220-74-62.

4.6.5 Cold bend. - The cable shall not be damaged after being conditioned for 20 hours at -40°C and then while still in the cold chamber being pulled over two 12 inch diameter rollers so as to impose an "S" bend.

4.6.6 Crack resistance. - The cable shall be resistant to stress cracking when tightly wrapped around a mandrel three inches in diameter for a continuous 24 hour period of time.

4.6.7 D.C. resistance. - The specimen shall consist of a length of coaxial cable at least 50 feet long which has been measured to an accuracy of 1 percent. The resistance shall be measured at a temperature of 20°C or corrected appropriately. The resistance shall be recorded in ohms per 1000 feet.

4.6.8 Resistance at 20 kHz. - The specimen shall consist of a straight length of coaxial cable at least 50 feet long measured to an accuracy of 1 percent. The center conductor shall be connected to the shield at one end. The resistance at 20 kHz of the center conductor and shield connected in series shall be measured with a suitable bridge or potentiometer to an accuracy of 1 percent. The resistance shall be recorded in ohms per 1000 feet of cable although that value will represent 2000 feet of the combined conductors. The resistance shall be measured at a temperature of 20°C or corrected appropriately.

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4.6.9 Attenuation. - The attenuation of the coaxial cable shall be measured at 2 MHz, 10 MHz, and 30 MHz in accordance with the procedure specified in MIL-C-17. The attenuation shall be calculated and recorded in dB per 100 feet to an accuracy of 10 percent.

4.6.10 Watertightness. - The cable core, up to and including the sheath of 3.3.5, shall be capable of withstanding a hydrostatic pressure of 600 psig applied to the open end of a 5-foot specimen employing dash no. 19 of USNUSL Test Gland Assembly Unit Number 44500B for a period of 2 hours with no leakage (see 3.3.4).

4.7 Inspection of preparation for delivery. - The inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Domestic shipment and early material use. -

5.1.1 Packaging. - Cable in the length specified (see 3.5) shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early use.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings Rules and Regulations or other carrier regulations as applicable to the mode of transportation. Reels shall have the following dimensions, hub not less than 18 inches diameter, width not more than 24 inches, and outside diameter not to exceed 36 inches.

5.1.3 Marking. - Shipment marking information shall be provided in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, specification number, contract or order number, date of manufacture (month and year), contractor's name, and destination.

5.1.3.1 Marking of each reel or spool shall be located on the flange area in a manner designed to preclude the possibility of the marking's becoming illegible during use.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.2.1.1 Packaging, packing, and marking. - Cable in lengths specified (see 3.5) shall be packaged level A or C as specified (see 6.1), packed level A or B as specified (see 6.1) and marked in accordance with the requirements of MIL-C-12000.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

6.1.1 Procurement requirements:

(a) Title, number, and date of this specification.
(b) Length if other than specified in 3.5.
(c) First article sample submitted if other than specified (see 4.4).
(d) Selection of applicable levels of packaging, packing, and marking (see 5.1 and 5.2).

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6.1.2 Contract data requirements. - Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

(a) Quality control system (see 4.1.1)
(b) First article inspection report (see 4.4)
(c) First article sample inspection report (see 4.4.1).

6.2 First article. - Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy-EC

(Project 6145-N112)

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NAVAL ELECTRONIC SYSTEMS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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